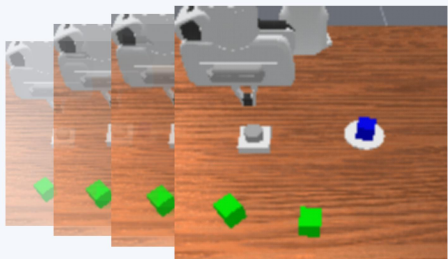


## a) Hierarchical Memory Update and Retrieval

### Hierarchical Memory System

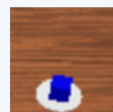
#### Short-Term Visual Memory



#### Semantic Event Memory



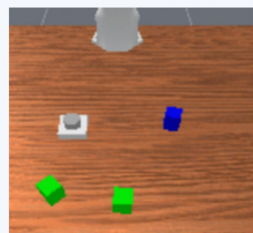
button  
pressed



blue cube  
highlighted

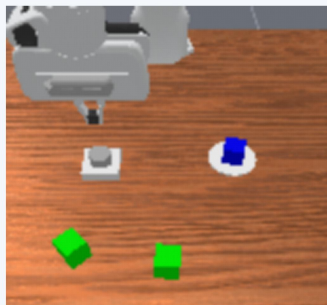
#### Executable Subtask List

1. press the button
2. pick up the blue cube



no future cues yet

$t_0$



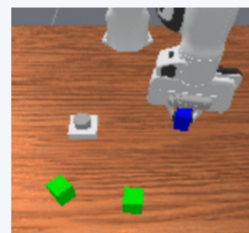
evidence is available

$t_1$

update subtask list

1. press the button

- ✓ 2. pick up the blue cube



execute the subtask

$t_2$

time axis



query  $P_t$  at  $t_2$   
and decide the  
next subtask

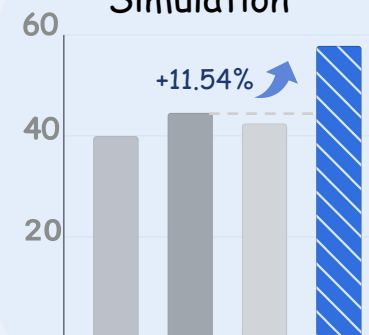
subtask



output action

## b) Results

### Simulation



### Real-world

